

Element C V1

MyDesign Portfolio

Element C Initial Template - AT - 18 points

****The requirements from Element C will be used for Elements D, E, F, and H.****

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The Arduino-based reaction time tester must measure and display user reaction time with a resolution of ± 5 milliseconds or better per trial using a consistent digital timing method.

This requirement is directly informed by the limitations identified in smartphone apps and ruler-drop tests (Element B). Both systems suffer from latency, human timing error, and low precision. Laboratory-grade systems (e.g., response box systems) demonstrate that millisecond-level accuracy is necessary for valid cognitive reaction measurement, so the Arduino system must approach similar precision to improve reliability.

The requirement was approved by the stakeholder after clarification that “precision” meant millisecond-level consistency rather than approximate timing. Feedback: “Make sure it’s more accurate than phone apps, ideally something we can trust for comparisons.” No revision required.

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Design Requirement #2 (HIGH Priority)

The device must randomly generate a stimulus delay between 2 and 8 seconds before activating a visual or auditory signal to prevent user anticipation.

Prior systems such as simple app testers and ruler-drop tests allow users to anticipate timing patterns, reducing validity of measured reaction time. Research-based systems like the Psychomotor Vigilance Task use randomized delays specifically to prevent predictive responses, ensuring that measured results reflect true reaction speed rather than expectation.

The requirement was approved. Feedback: "Random timing is important so users don't cheat or guess when it will happen."

Design Requirement #3 (HIGH Priority)

The system must store and calculate the average reaction time across trials per session, displaying both individual results and the final mean.



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Single-trial reaction tests (e.g., ruler drop or basic apps) are unreliable due to variability in human motor response. Research systems like the PVT use multiple trials to improve statistical reliability. Averaging multiple attempts reduces random error and provides a more valid measurement of true reaction speed.

The requirement was approved. Feedback: "I want to see improvement over time, not just one result."

Design Requirement #4 (HIGH Priority)

The device must provide a clear visual stimulus (LED) and optional audible stimulus (buzzer) with response input recorded via a physical push button.

Prior smartphone-based systems are affected by touchscreen latency and input lag. Laboratory response boxes and sports training systems show that dedicated physical input hardware improves timing accuracy and reduces variability. Using Arduino-controlled LED/buzzer output and a button ensures a controlled stimulus-response loop.

The requirement was approved. Feedback: "A button is better than a screen because it feels more precise and consistent."



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Design Requirement #5 (HIGH Priority)

The system must display reaction time results. The system must provide a visual confirmation of each trial outcome using a 16x2 LCD screen or serial monitor, displaying reaction time in milliseconds within 500 ms of user response input. It is on a real-time output screen (LCD or serial monitor) immediately after each trial (within 1 second of response).

This revision improves precision and measurability compared to generic “real-time feedback.” Prior systems such as laboratory response boxes and PVT setups emphasize immediate result feedback to ensure usability and accurate trial separation. Unlike smartphone apps, which may introduce display lag or inconsistent UI refresh rates, Arduino-based serial or LCD output allows deterministic timing control, ensuring feedback is tightly coupled to the recorded response event.

The requirement was disapproved by the stakeholder. Feedback: “I need less error 500 milliseconds is too much”

Design Requirement #6 (LOW Priority)

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The system should be powered via USB or portable battery supply and remain fully functional without requiring a constant computer connect

Many laboratory-grade systems require fixed setups, limiting accessibility. Prior design analysis showed portability is a weakness in professional systems. A standalone Arduino system improves usability in classrooms, sports environments, or home testing scenarios.

The requirement was approved. Feedback: "It should be easy to move around and use anywhere."

Design Requirement #7 (LOW Priority)

The device should include a score tracking feature that records the fastest reaction time achieved in a session.

Sports reaction training systems often highlight peak performance to motivate users. While not essential for measurement accuracy, this feature increases engagement and



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encourages repeated testing, improving dataset size and reliability.

The requirement was not approved.
Feedback: "The high score does not matter, it is the average time that should be improved"

Rubric

Element C: Presentation and Justification of Solution Design Requirements						
C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	No
C2. Requirements list. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are listed, prioritized, clear, and detailed	are listed, prioritized, generally clear, and detailed	are listed, prioritized, generally clear, and somewhat detailed	are listed and prioritized, but some or all may be incomplete and/or lack specificity	are listed, formatted, and prioritization is attempted, but these may be partial or overly general	are not present or are too vague
C3. Quantity of high-priority design requirements.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	Five	Four	Three	Two	One	None

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_____ of the design requirements listed are identified as high priority.						
C4. Objective & measurable. Design requirements _____.	score = 5 are consistently objective and measurable	score = 4 are nearly always objective and measurable	score = 3 are generally objective and measurable	score = 2 may be only sometimes objective and/or measurable	score = 1 are insufficiently measurable	score = 0 are not measurable
C5. Leading to solution. Design requirements _____.	score = 5 would be highly likely to lead to a tangible and viable problem solution	score = 4 would be likely to lead to a tangible and viable problem solution	score = 3 has the potential to lead to a tangible and viable problem solution	score = 2 has some potential to lead to a tangible and viable problem solution	score = 1 have little potential to lead to a tangible and viable problem solution	score = 0 cannot lead to a tangible and viable problem solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of	score = 0 no primary stakeholder groups.